

Python Training Program Documentation

Module 1 – Introduction to Python

Objective

Module 1 is designed to build a strong foundation in Python programming. Students will learn basic programming concepts, problem-solving techniques, and gain hands-on coding experience through notebooks, assignments, and coding practice platforms.

1. Module 1 Notebook – Introduction to Python

Notebook Link:

https://colab.research.google.com/drive/157PLrHeGzP09j60pMuk_hArm5khxRx5Y

Description:

This notebook introduces learners to Python programming fundamentals including:

- Variables and Data Types
- Input and Output Operations
- Operators
- Conditional Statements
- Loops
- Functions
- Basic Programming Concepts
- Lists, Tuples, Sets, and Dictionaries
- String Manipulation
- Basic Problem Solving

2. Python Assignment 1

Document Link:

https://docs.google.com/document/d/1sd5qqggT0VWI-_xdrGMS1ZEH9QczZ-dCIYsOUSdWH2g/edit?tab=t.0#heading=h.hlar8zxnyqhj

Description:

Assignment designed to evaluate students on:

- Basic Python Syntax
 - Variables and Operators
 - Conditional Statements
 - Loops
 - Functions
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3. Python Assignment 2

Document Link:

https://docs.google.com/document/d/1oKidScc6oovFJaxtIWFgdb6pk2LrpBvS_IDGoXCvYtA/edit?tab=t.0#heading=h.f6x17cgh8vw9

Description:

Assignment covering intermediate Python concepts such as:

- Lists, Tuples, Sets, and Dictionaries
 - Functions
 - Problem Solving
 - Data Manipulation
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4. Python Assignment 3

Document Link:

<https://docs.google.com/document/d/1lEXWJ3kAhhBEopnZxvqFMw5EGy-B6JAp21wX9Umjm-4/edit?tab=t.0#heading=h.eu8jpxntx6e>

Description:

Advanced assignment focusing on:

- Python Programming Logic
 - Problem Solving
 - Functions and Data Structures
 - Real-world Programming Exercises
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Mandatory Coding Practice Platforms

Students are required to create accounts on the following platforms to improve coding skills, logical thinking, and problem-solving abilities.

5. SoloLearn (Mandatory)

Website:

<https://www.sololearn.com/>

Importance:

- SoloLearn is a beginner-friendly learning platform.
 - It provides interactive Python lessons with quizzes and exercises.
 - Students can practice coding directly on mobile devices.
 - SoloLearn is available as both a website and a mobile application (Android and iOS), making learning possible anytime and anywhere.
 - Students are encouraged to complete the Python course and earn certificates.
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6. HackerRank (Mandatory)

Website:

<https://www.hackerrank.com/>

Importance:

- Improves problem-solving and coding skills.
- Provides real-world coding challenges.
- Widely used by companies during technical hiring processes.

- Helps students prepare for placement tests and coding interviews.
 - Students should regularly solve Python programming challenges.
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7. LeetCode (Recommended)

Website:

<https://leetcode.com/>

Importance:

- Industry-standard platform for coding interview preparation.
 - Enhances logical thinking and algorithmic problem-solving skills.
 - Frequently used by top technology companies for technical interviews.
 - Recommended for students aiming for software development careers.
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8. CodeChef (Recommended)

Website:

<https://www.codechef.com/>

Importance:

- Excellent platform for competitive programming.
 - Helps students improve coding speed and accuracy.
 - Organizes regular coding contests and competitions.
 - Encourages analytical and problem-solving skills.
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9. GeeksforGeeks (Mandatory)

Website:

<https://www.geeksforgeeks.org/>

Importance:

- Comprehensive resource for programming concepts.
- Contains tutorials, interview questions, articles, and coding problems.
- Useful for revision and self-learning.

- Provides extensive Python learning materials.
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10. GitHub (Mandatory)

Website:

<https://github.com/>

Importance:

- Industry-standard platform for version control and code sharing.
 - Students should upload assignments and projects to GitHub.
 - Helps in building a professional portfolio.
 - Essential skill for software developers and data professionals.
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Recommended Learning Practice

Students are advised to:

- Practice coding for at least 30–60 minutes every day.
 - Complete all notebook exercises and assignments.
 - Solve a minimum of 5 coding problems per week on HackerRank or LeetCode.
 - Maintain a GitHub repository for all assignments and projects.
 - Participate in coding contests and challenges whenever possible.
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OneDrive Class Notes

Link:

<https://1drv.ms/o/c/350d1dd72dbeac65/IqDi5Nr3-zYyTK3X7UAP8yjPAZevZVF53w6RmvpPb7j6YG4?e=2AITdr>

Description:

Central repository containing:

- Learning Materials

- Class Notes
- Datasets
- Assignments
- Additional Practice Resources
- Supporting Documents for the Training Program

Students are expected to regularly check the repository for updates and newly added learning resources.